

Running Identification of Fixed Equity Strategies

Alexandros E. Tzikas

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Abstract

We study whether a trading strategy can be identified from its return sequence. Using four candidate strategies of equities, we compare daily inverse-error weights, covariance matching, and variance-weighted PCA. Across twelve fixed black-box strategies, with each being a variant of one of the four candidates, correct-identification rates are 76.4%, 76.1%, and 92.5%, respectively.

1 Experiment and data

We use January 3, 2006–September 14, 2026 and a fixed universe of 260 stocks: January 2006 S&P 500 constituents with complete data and a 252-session warmup. Index constituents are from [1]; adjusted closes and volumes are from Yahoo Finance [2]. Later index additions and deletions do not change the universes. Data from 2004–2005 initialize signals.

Candidates rebalance every five trading days. *Momentum* equally weights the top half of its universe by 20-day returns; *mean reversion* uses the bottom half by five-day returns. *Risk parity* equalizes risk contributions using the full trailing 60-day covariance with 10% diagonal shrinkage. *Equal weight* holds every stock in its universe equally. Portfolios are long-only, with two basis points of cost per unit traded, zero cash interest, and weights that drift between rebalances.

We construct three black-box strategies from each candidate. We therefore have three variants each of momentum, mean reversion, risk parity, and equal weight. Their parameters are shown below. Profile A uses the same 260 stocks as the candidates; B and C use fixed subsets of 200 and 100 stocks, selected by average dollar volume over the last 20 sessions of 2005. Each black-box strategy keeps its parameters, rebalance schedule, and stock universe throughout the experiment.

Profile	MOM/MR/RP lookbacks	Rebalance	Selected fraction
A	10/2/40	1 day	1/2
B	60/10/120	10 days	1/4
C	120/20/252	21 days	1/5

Lookbacks and rebalance intervals are in trading days. The selected fraction applies only to momentum and mean reversion; risk parity and equal weight use their whole fixed universe. Equal weight has no signal lookback.

2 Daily running identification

We aim to identify the candidate of which a black-box strategy is a variant, using only returns. For example, a momentum variant should match the momentum candidate. At date t , let T be the number of observed dates, x_{sj} candidate j 's return, and y_s the black-box return at any observed time s before or at t . Observations through that day's close receive equal weight. Form $Z \in \mathbb{R}^{5 \times T}$ with the four candidate histories and the black-box history as rows. Demean each row to obtain $\tilde{Z}$; do not scale by volatility.

Squared-error weights and SSE. Choose the smallest S_j or largest p_j :

$$S_j = \sum_{s=1}^T (y_s - x_{sj})^2, \quad L_j = S_j/T, \quad p_j = \frac{(\epsilon + L_j)^{-1}}{\sum_{k=1}^4 (\epsilon + L_k)^{-1}}. \quad (1)$$

Set $\epsilon = (0.0001)^2$. These two rules give identical leading candidates.

Empirical covariance. Set $\hat{\Sigma} = \tilde{Z}\tilde{Z}^T/T$ and choose the candidate minimizing

$$d_j = \hat{\Sigma}_{55} + \hat{\Sigma}_{jj} - 2\hat{\Sigma}_{j5} = \widehat{\text{Var}}(y - x_j). \quad (2)$$

Variance-weighted PCA projection. Write $\tilde{Z} = U\Lambda V^T$. Project each row onto all temporal components, $Q = \tilde{Z}V = U\Lambda$. Weight squared projection differences by each component's explained-variance share:

$$w_k = \frac{\lambda_k}{\sum_{\ell=1}^5 \lambda_\ell}, \quad \hat{j}_{\text{PCA}} = \arg \min_{j \leq 4} \sum_{k=1}^5 w_k (Q_{5k} - Q_{jk})^2. \quad (3)$$

Here $\lambda_k = \Lambda_{kk}^2/T$; zero-variance components have zero weight. The score is invariant to signs and repeated-eigenspace rotations. Covariance and PCA are undefined on day one; PCA is also undefined at zero total variance.

Family	Profile	Weights (%)	Covariance (%)	PCA (%)
MOM	A	99.1	99.1	98.3
MOM	B	99.1	99.2	88.8
MOM	C	4.9	5.0	39.7
MR	A	0.1	0.1	96.2
MR	B	92.6	89.9	99.9
MR	C	25.9	24.7	99.5
RP	A	99.9	99.9	99.9
RP	B	99.9	99.6	99.8
RP	C	95.4	95.4	90.5
EW	A	100.0	100.0	100.0
EW	B	100.0	100.0	99.8
EW	C	99.8	99.8	97.2

Table 1. Daily correct-family fractions on defined dates; ties split credit. MOM: momentum; MR: mean reversion; RP: risk parity; EW: equal weight. Final correct counts are 9/12, 9/12, and 12/12, respectively. All three methods score 100% on exact-copy controls.

3 Return paths and daily identification

Weighted PCA achieves 92.5% correct-family identification, compared with 76.4% for error weights and 76.1% for covariance matching. It correctly identifies all 12 black boxes on the final date; the other methods identify 9 of twelve. The largest gains occur in mean reversion: profile A improves from 0.1% with error weights to 96.2% with PCA, and profile C from 25.9% to 99.5%.

Replacing absolute projection differences with squared differences improves PCA accuracy from 91.36% to 92.47%, a gain of 1.11 percentage points. The largest gain is equal-weight profile C, from 87.7% to 97.2%; mean-reversion profile A falls slightly, from 97.5% to 96.2%. Both PCA versions identify all twelve strategies correctly on the final date.

PCA’s variance weights emphasize dominant shared return patterns and reduce the influence of low-variance components. This appears to reduce sensitivity to parameter-specific tracking differences, consistent with the large gains for mean-reversion black boxes.

Figures 1–2 show compounded net wealth on logarithmic axes. Black boxes within the same family can follow different wealth paths because their lookbacks, stock subsets, and rebalancing intervals differ, even though each rule remains fixed throughout the experiment.

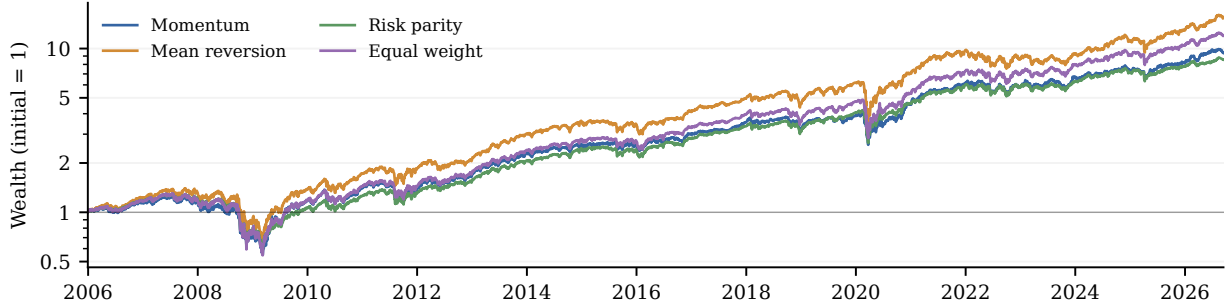


Figure 1. Four canonical strategies: growth of 1 unit of capital, logarithmic axis.

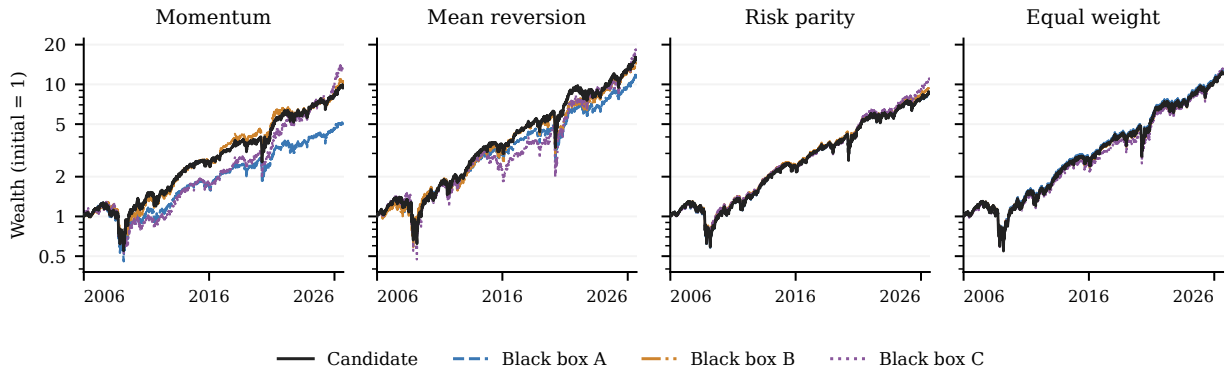


Figure 2. Twelve black boxes (A–C) and their family candidates. Shared logarithmic axes show 1 plus cumulative net return, starting from wealth 1.

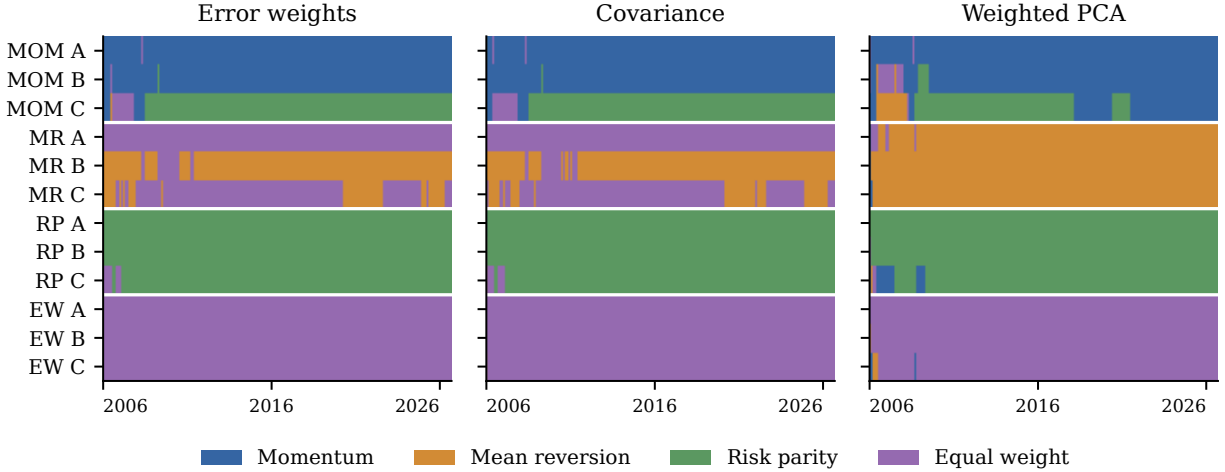


Figure 3. Daily leaders by true family and profile; colors identify the selected candidate.

Figure 3 shows when each method selects the correct family and when it confuses one family with another. Labels can change as return observations accumulate; the generating family and its parameters do not change. Error weights and covariance matching agree on 99.6% of defined dates, whereas PCA separates several mean-reversion profiles that those methods label as equal weight.

PCA’s advantage is not uniform. For equal-weight profile C, its accuracy is 97.2%, below 99.8% for error weights; momentum profile C remains difficult for all methods. The comparison therefore supports better average identification by weighted PCA in this experiment, while retaining meaningful failures at the individual-profile level. All methods recover the exact-copy controls on every defined date.

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References

- [1] fja05680 and contributors. Historical S&P 500 components and changes. github.com/fja05680/sp500. Data snapshot retrieved September 15, 2026 (New York time).
- [2] Yahoo Finance. Historical adjusted-close and volume data, finance.yahoo.com.